

Chapter 2 Fundamental equations

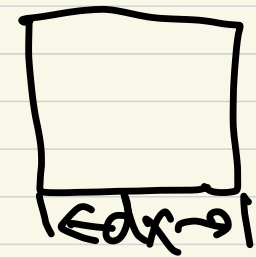
- Three basic equations

- Conservation of mass (continuity)
- Conservation of momentum (Newton's 2nd law)
- Conservation of energy (1st law of thermodynamics)

- All the three conservation laws are Lagrangian in nature.

- For mass $m = \rho V$

$$\frac{Dm}{Dt} = 0 = \frac{D}{Dt}(\rho V) = \frac{D\rho}{Dt} V + \rho \frac{DV}{Dt}$$



$$u + \frac{\partial u}{\partial x} dx$$



$$\rho \frac{\partial u}{\partial x} dx \cdot dt$$

$$\text{Change of } \rho V = \frac{\partial \rho}{\partial x} dx \, dt \, dy \, dz$$

$$\Delta \psi = \frac{\partial \psi}{\partial x} \Delta x \cdot \psi$$

$$\psi = dx dy dz$$

$$\frac{Dm}{Dt} = \rho \frac{D\psi}{Dt} + \psi \frac{D\rho}{Dt}$$

$$\left[\frac{1}{\psi} \frac{D\psi}{Dt} = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y} + \frac{\partial \psi}{\partial z}, \quad \nabla \cdot \vec{v} \right] \rho \psi$$

$$= \rho \psi \nabla \cdot \vec{v} + \psi \left(\frac{D\rho}{Dt} + \nabla \cdot \nabla P \right)$$

$$= \psi \left[\frac{D\rho}{Dt} + \nabla \cdot \nabla P + \rho \nabla \cdot \vec{v} \right]$$

$$= \psi \left[\frac{D\rho}{Dt} + \nabla \cdot (\rho \vec{v}) \right] = 0$$

$$\therefore \frac{D\rho}{Dt} + \nabla \cdot (\rho \vec{v}) = 0$$

$$\frac{D\rho}{Dt} + \frac{d}{dx_j} (\rho u_j) = 0$$

- For incompressible flow $\rho = C$

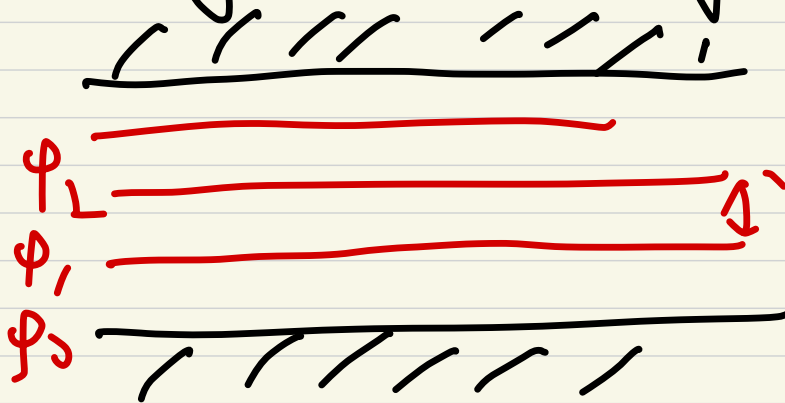
$$\nabla \cdot \vec{v} = 0$$

$$\frac{\partial u_j}{\partial x_j} = 0$$

- 2D flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

- Define stream function



$$\psi_2 - \psi_1 = \frac{\dot{m}}{\rho}$$

$$u = \frac{\partial \psi}{\partial y} \quad \longleftrightarrow \quad dy = \frac{1}{u} dy$$

$$v = -\frac{\partial \psi}{\partial x} \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\rho \frac{\partial^2 \psi}{\partial x^2 \partial y} - \rho \frac{\partial^2 \psi}{\partial x \partial y^2} = 0$$

- The no. of dependent variable reduced by one.

- The penalty: derivatives are increased by one order.

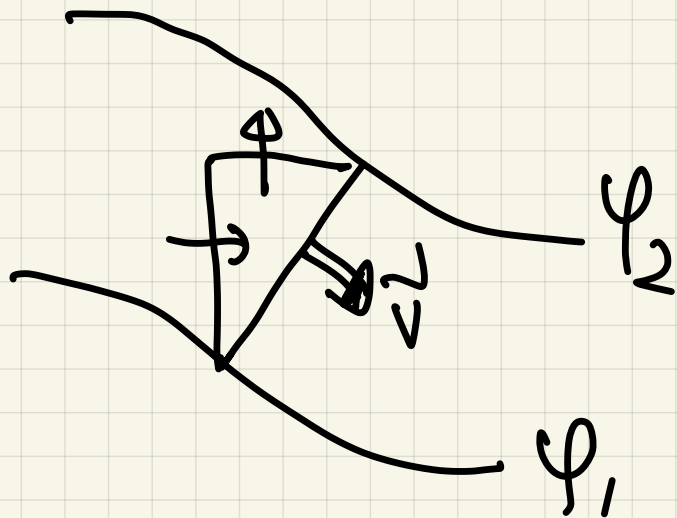
- Vorticity $\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

$$-\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \omega_z \Rightarrow \boxed{\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega_z}$$

- The physical meaning of stream function

$$dy = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v dx + u dy$$

$$= \vec{p} \vec{v} \cdot d\vec{A} = d\psi$$



- For constant ψ ,
 $d\psi = 0$, there is
 no mass flow rate.

$\psi_2 - \psi_1 =$ mass flow rate between
 the streamlines.

- Conservation of momentum.

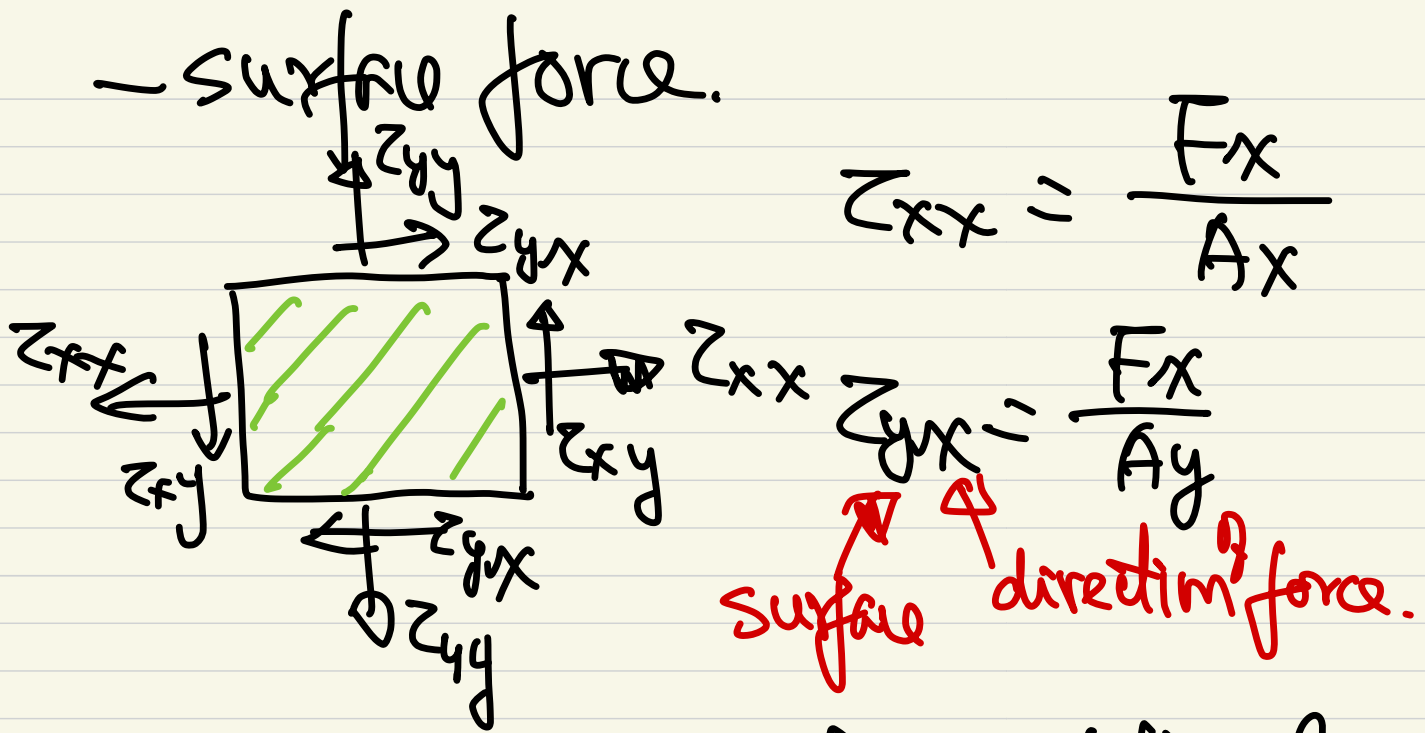
• Newton's 2nd law.

$$m \vec{a} = \sum \vec{F}, \quad m = \rho V$$

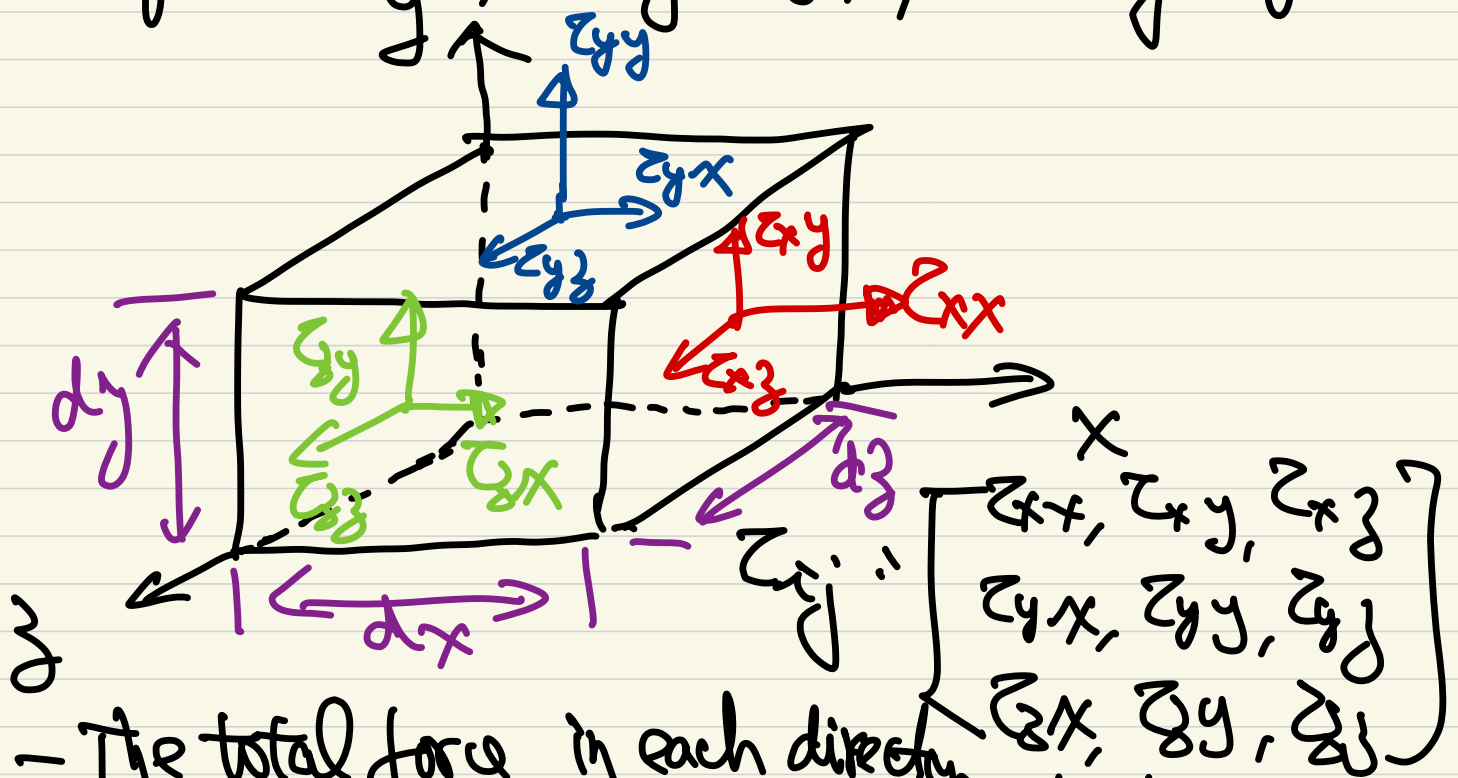
$$\rho \frac{d\vec{v}}{dt} = \sum \vec{f} = \vec{f}_{body} + \vec{f}_{surface}.$$

— per unit volume.

- Surface force.



- In order to maintain rotational equilibrium, $\tau_{xy} = \tau_{yx}$, i.e. $\tau_{ij} = \tau_{ji}$



- The total force in each direction

$$\begin{cases} dF_x = \tau_{xx} dy dz + \tau_{yx} dx dz + \tau_{zx} dx dy \\ dF_y = \tau_{xy} dy dz + \tau_{yy} dx dz + \tau_{zy} dx dy \\ dF_z = \tau_{xz} dy dz + \tau_{yz} dx dz + \tau_{zz} dx dy \end{cases}$$

- If the element is accelerating,

$$\tau_{xx}(x+dx) = \tau_{xx}(x) + \frac{\partial \tau_{xx}}{\partial x} dx$$

$$\tau_{yx}(y+dy) = \tau_{yx}(y) + \frac{\partial \tau_{yx}}{\partial y} dy$$

$$\tau_{zx}(z+dz) = \tau_{zx}(z) + \frac{\partial \tau_{zx}}{\partial z} dz$$

- Hence, the net force on the element in x direction.

$$dF_{x, \text{net}} = \frac{\partial \tau_{xx}}{\partial x} dx dy dz + \frac{\partial \tau_{yx}}{\partial y} dy dx dz + \frac{\partial \tau_{zx}}{\partial z} dz dx dy$$

- on a unit volume.

$$\frac{dF_{x, \text{net}}}{V} = f_x = \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}$$

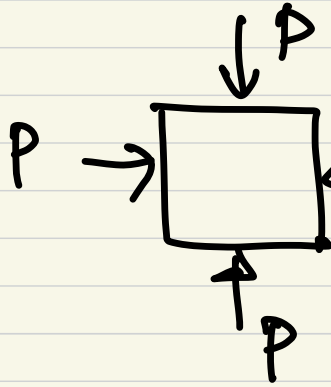
i.e. $f_{\text{surface}} = \nabla \cdot \tau = \frac{\partial \tau_{ij}}{\partial x_j}$

- Newton's 2nd law

$$\rho \frac{dV}{dt} = \rho g + \nabla \cdot \tau$$

τ_{ij} is yet to be determined.

- Fluid at rest, the viscous stress must vanish.



- The shear stresses are zero.

- The normal stresses equal the hydrostatic pressure.

$$\tau_{xx} = \tau_{yy} = \tau_{zz} = -P, \quad \tau_{ij} = 0, \quad i \neq j$$

$$\tau_{ij} = -P\delta_{ij} + \tau_{ij}^*$$

- Stokes three postulates (1845).

1. The fluid is continuous, and τ_{ij} is at most a linear function of ϵ_{ij} ($\epsilon_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i})$)
2. The fluid is isotropic, i.e., its properties are independent of directions. Therefore the conservation law is independent of the coordinate axes in which it is expressed.
3. When the strain rate is zero the deformation law must reduce to the hydrostatic pressure i.e., $\tau_{ij} = -P\delta_{ij} \Leftrightarrow \epsilon_{ij} = 0$.

- For isotropic condition, the principal stress axes are identical with the principal strain rate axes.
- Let x, y, z be the principal axes, where shear stress and shear strain rate vanish.

$$\epsilon_{ij} = \begin{bmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_2 & 0 \\ 0 & 0 & \epsilon_3 \end{bmatrix}, \quad \tau_{ij} = \begin{bmatrix} \tau_{11} & 0 & 0 \\ 0 & \tau_{22} & 0 \\ 0 & 0 & \tau_{33} \end{bmatrix}$$

- τ_{ij} is a linear function of ϵ_{ij} .

Thus,

$$\begin{aligned} \tau_{11} &= -p + C_1 \epsilon_1 + C_2 \epsilon_2 + C_3 \epsilon_3 \\ &= -p + (C_1 - C_2) \epsilon_1 + C_2 (\epsilon_1 + \epsilon_2 + \epsilon_3) \\ &= -p + K \epsilon_1 + C_2 \nabla \cdot \mathbf{v} \end{aligned}$$

$C_2 = C_3$ is for isotropic condition

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$C_1 = C_{11} = \frac{\partial \tau}{\partial \epsilon_1}$$

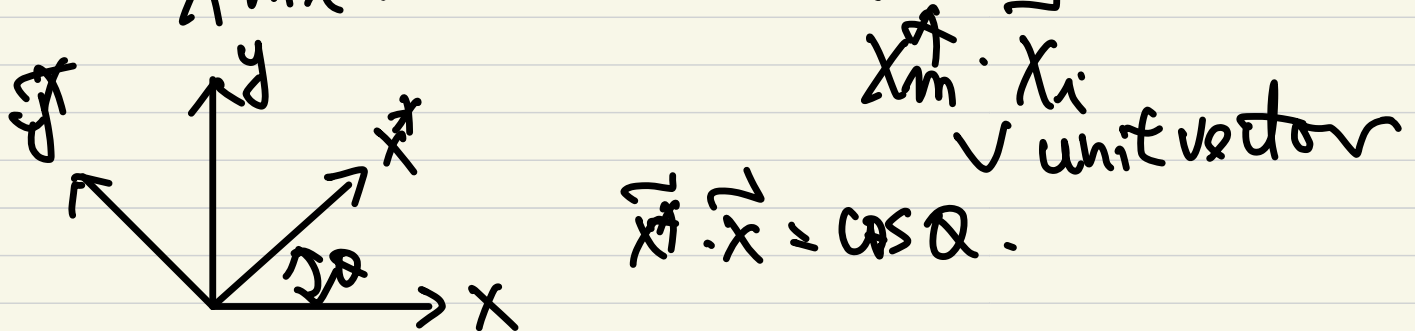
$$C_2 = \frac{\partial \tau}{\partial \epsilon_2}$$

$$C_3 = \frac{\partial \tau}{\partial \epsilon_3}$$

- Transform the stresses to coordinate x^*, y^*, z^* .

$$\tau_{mn}^* = \lambda_{mi} \lambda_{nj} \tau_{ij} \quad \text{at principal axes.}$$

λ_{mi} = direction cosine. j^{th}



$\tau_{ij} \rightarrow \tau_{ji}$ at the principal axes.

$$\lambda_{x^*x}^2 + \lambda_{x^*y}^2 + \lambda_{x^*z}^2 = 1$$

$$\tau_{x^*x^*} = \lambda_{x^*x}^2 \tau_{xx} + \lambda_{x^*y}^2 \tau_{yy} + \lambda_{x^*z}^2 \tau_{zz}$$

$$\sigma_{x^*x^*} = \lambda_{x^*x}^2 \sigma_1 + \lambda_{x^*y}^2 \sigma_2 + \lambda_{x^*z}^2 \sigma_3$$

$$\begin{cases} \tau_{xx} = -p + k \sigma_1 + C_2 \nabla \cdot \vec{v} \\ \tau_{yy} = -p + k \sigma_2 + C_2 \nabla \cdot \vec{v} \\ \tau_{zz} = -p + k \sigma_3 + C_2 \nabla \cdot \vec{v} \end{cases}$$

$$\begin{aligned}
Z_{xxx} &= \lambda_{xx}^2 (-p + K \epsilon_1 + C_2 \vec{\sigma} \cdot \vec{v}) \\
&+ \lambda_{xy}^2 (-p + K \epsilon_2 + C_2 \vec{\sigma} \cdot \vec{v}) \\
&+ \lambda_{xz}^2 (-p + K \epsilon_3 + C_2 \vec{\sigma} \cdot \vec{v}) \\
&= (-p + C_2 \vec{\sigma} \cdot \vec{v}) (\lambda_{xx}^2 + \lambda_{xy}^2 + \lambda_{xz}^2) \\
&+ K (\lambda_{xx}^2 \epsilon_1 + \lambda_{xy}^2 \epsilon_2 + \lambda_{xz}^2 \epsilon_3) \\
&= -p + K \epsilon_{xxx} + C_2 \vec{\sigma} \cdot \vec{v}
\end{aligned}$$

$$\begin{aligned}
Z_{xyx} &= \lambda_{xx} \lambda_{yx} Z_{xx} + \lambda_{xy} \lambda_{yy} Z_{yy} \\
&+ \lambda_{xz} \lambda_{yz} Z_{zz}
\end{aligned}$$

$$\begin{aligned}
\epsilon_{xyx} &= \lambda_{xx} \lambda_{yx} \epsilon_1 + \lambda_{xy} \lambda_{yy} \epsilon_2 \\
&+ \lambda_{xz} \lambda_{yz} \epsilon_3
\end{aligned}$$

$$\begin{cases}
Z_{xx} = -p + K \epsilon_1 + C_2 \vec{\sigma} \cdot \vec{v} \\
Z_{yy} = -p + K \epsilon_2 + C_2 \vec{\sigma} \cdot \vec{v} \\
Z_{zz} = -p + K \epsilon_3 + C_2 \vec{\sigma} \cdot \vec{v}
\end{cases}$$

$$\begin{aligned}
 \tau_{xy}^* &= \lambda_{yx}^* \lambda_{yx}^* (-p + K \epsilon_1 + C_2 \nabla \cdot \vec{v}) \\
 &\quad + \lambda_{xy}^* \lambda_{xy}^* (-p + K \epsilon_2 + C_2 \nabla \cdot \vec{v}) \\
 &\quad + \lambda_{yz}^* \lambda_{yz}^* (-p + K \epsilon_3 + C_2 \nabla \cdot \vec{v}) \\
 &= (-p + C_2 \nabla \cdot \vec{v}) \left(\lambda_{yx}^* \lambda_{yx}^* + \lambda_{xy}^* \lambda_{xy}^* \right. \\
 &\quad \left. + \lambda_{yz}^* \lambda_{yz}^* \right) \\
 &\quad + \lambda_{yx}^* \lambda_{yx}^* K \epsilon_1 + \lambda_{xy}^* \lambda_{xy}^* K \epsilon_2 \\
 &\quad + \lambda_{yz}^* \lambda_{yz}^* K \epsilon_3 \\
 &= K \epsilon_{xy}^*
 \end{aligned}$$

- For Newtonian fluid

$$\tau_{xy} = 2\mu \epsilon_{ij} = 2\mu \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)$$

$$\underline{\underline{K = 2\mu}}$$

- Thus,

$$\tau_{ij} = -p \delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k}$$

- Define the mechanical pressure. τ_{ij}^*

$$\bar{p} = -\frac{1}{3} (\tau_{xx} + \tau_{yy} + \tau_{zz})$$

$$= -\frac{1}{3} \left(-p + 2\mu \frac{\partial u}{\partial x} + \lambda \frac{\partial u_k}{\partial x_k} \right.$$

$$\left. - p + 2\mu \frac{\partial v}{\partial y} + \lambda \frac{\partial u_k}{\partial x_k} \right.$$

$$\left. - p + 2\mu \frac{\partial w}{\partial z} + \lambda \frac{\partial u_k}{\partial x_k} \right]$$

$$= p - \left(\lambda + \frac{2}{3}\mu \right) \nabla \cdot \vec{u}$$

thermodynamic pressure

- Stokes hypothesis

$$\lambda + \frac{2}{3}\mu = 0 \Rightarrow \lambda = -\frac{2}{3}\mu.$$

- Navier-Stokes equation

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = - \frac{\partial p}{\partial x_i}$$

$$+ \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \delta_{ij} \frac{\partial u_k}{\partial x_k} \right]$$

- For incompressible, $\rho = c$.

$$\nabla \cdot \vec{u} = \frac{\partial u_k}{\partial x_k} = 0$$

$$\rho \frac{\partial u_i}{\partial t} + \rho u_j \frac{\partial u_i}{\partial x_j} = - \frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j^2}$$

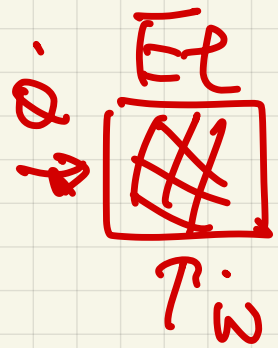
- Energy equation.

$$e_t = e + \frac{v^2}{2} - g \cdot r$$

↑
internal
energy

↑
kinetic
energy

↑
potential
energy



$$E_t = m e_t = m \left(e + \frac{v^2}{2} - g \cdot r \right), \quad m = \rho V.$$

$$\frac{DE_t}{dt} = \frac{Dme_t}{dt} = e_t \frac{Dm}{dt} + m \frac{De_t}{dt}$$

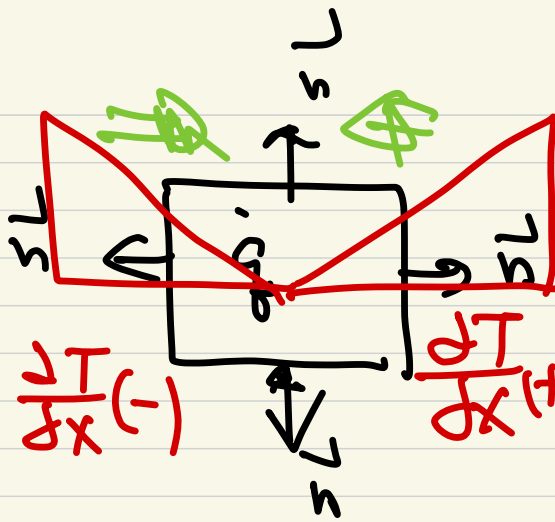
$$= \underset{\rho V}{m} \left(\frac{De}{dt} + v \cdot \left(\frac{Dv}{dt} - g \right) \right) = \boxed{\dot{Q} + \dot{W}}$$

N-S eq

$$\left[\rho \frac{Dv}{dt} = \rho g + \sigma \cdot \nabla v \right] \Rightarrow \rho \left[\frac{Dv}{dt} - g = \sigma \cdot \nabla v \right]$$

- work $\nabla v \cdot v = (\nabla \cdot v) \cdot v$

$$\dot{W} = \int_{CS} v \cdot (\nabla \cdot v) dA = \int_{CV} \nabla \cdot (v \cdot v) dV.$$



$$\dot{Q} = \int_{\omega} \rho \dot{q} d\omega + \int_{\kappa_s} R \nabla T \cdot \vec{n} dA$$

$$= \int_{\omega} \rho \dot{q} d\omega + \int_{\omega} (\nabla \cdot R \nabla T) d\omega$$

$$m = \rho \omega \Rightarrow \frac{D E}{D t} / \omega$$

$$\rho \frac{D \rho}{D t} + \nabla \cdot \left(\rho \frac{D \vec{v}}{D t} - \rho \vec{g} \right) = \rho \dot{q} + \nabla \cdot R \nabla T + \nabla \cdot (\vec{\tau} \cdot \vec{v})$$

$$\rho \frac{D \rho}{D t} = \rho \dot{q} + \nabla \cdot R \nabla T + \vec{\tau} \cdot \nabla \vec{v}$$

$$\tau_{ij} = -p \delta_{ij} + \mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \lambda \delta_{ij} \frac{\partial v_k}{\partial x_k}$$

$$z_j \frac{\partial u}{\partial x_j} = [-p \delta_{ij} + \underbrace{\mu \left(\frac{\partial u}{\partial x_j} + \frac{\partial u}{\partial x_i} \right) + \lambda \delta_{ij} \frac{\partial u}{\partial x_k}}_{z_{ij}^{*}}] \frac{\partial u}{\partial x_j}$$

$$= -p \nabla \cdot \vec{v} + z_{ij}^{*} \frac{\partial u}{\partial x_j}$$

$$= -p \nabla \cdot \vec{v} + \vec{v} \cdot \nabla p$$

$$\frac{dp}{dt} + \nabla \cdot p \vec{v} = \underbrace{\frac{dp}{dt} + \vec{v} \cdot \nabla p}_{=0} + \underbrace{p \nabla \cdot \vec{v}}_{=0} = 0$$

$$\frac{dp}{dt} = \frac{dp}{dt} + \vec{v} \cdot \nabla p = -p \nabla \cdot \vec{v}$$

$$-p \nabla \cdot \vec{v} = \frac{dp}{dt} \Rightarrow -\nabla \cdot \vec{v} = \frac{1}{p} \frac{dp}{dt}$$

$$-p \nabla \cdot \vec{v} = \frac{p}{p} \frac{dp}{dt} = -p \frac{D}{Dt} \left(\frac{p}{p} \right) + \frac{dp}{dt}$$

$$p \frac{De}{Dt} + p \frac{D}{Dt} \left(\frac{p}{p} \right) = p \dot{\epsilon} + \nabla \cdot p \vec{v} + \vec{v} \cdot \nabla p$$

$$h = e + \frac{p}{\rho}$$

viscous dissipation

$$p \frac{Dh}{Dt} = \nabla \cdot p \vec{v} + p \dot{\epsilon} + \underbrace{\vec{v} \cdot \nabla p}_{=0} + \frac{dp}{dt}$$

$$h = C_p T$$

$$\rho \left(\frac{\partial C_p T}{\partial t} + u_j \frac{\partial C_p T}{\partial x_j} \right) = \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \rho \dot{q}$$

$$\tau_{ij}^* = \boxed{\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)} + \frac{\rho \mu}{2} \frac{\partial u_k}{\partial x_k} \frac{\partial u_l}{\partial x_l}$$

$$\tau_{ij}^* = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k}$$

- Incompressible flow: viscous dissipation

$$\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \frac{\partial u_i}{\partial x_j}$$

$$= 2\mu \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + 2\mu \frac{\partial v}{\partial y} \frac{\partial v}{\partial y}$$

$$+ \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \frac{\partial u}{\partial y} + \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \frac{\partial v}{\partial x}$$

$$= 2\mu \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right] + \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 > 0$$

- If viscous dissipation is negligible,
 $\rho = C, c_p = C, \dot{q} = 0, R = C.$

$$\frac{\partial T}{\partial t} + u_j \frac{\partial T}{\partial x_j} = \boxed{\frac{R}{\rho c_p}} \frac{\partial^2 T}{\partial x_j^2} \quad \text{A}$$

α : diffusivity

2-9 Non-dimensionalizing the basic eqn.

1. reference velocity v_0 , or $\frac{\mu_0}{\rho_0 L}$ if there is no free stream (natural convection)

2. Other free stream properties

P_0, ρ_0, T_0, μ_0

3. reference length L (body length or duct diameter)

4. residence time L/v_0 (steady flow)

ω^{-1} unsteady oscillating frequency

$\omega L/v_0$ - stroked number $\frac{\text{residence}}{\text{oscillation time}}$

- Dimensionless variables.

$$x^* = \frac{x}{L}, \quad p^* = \frac{p - p_0}{p_0 v_0^2}, \quad \rho^* = \frac{\rho}{\rho_0}$$

$$v^* = \frac{v}{v_0}, \quad \tau^* = \frac{T - T_0}{T_\infty - T_0}, \quad T^* = L \cdot \nabla = \frac{d}{d\left(\frac{x_i}{L}\right)}$$

$$t^* = \frac{t}{L/v_0}, \quad \mu^* = \frac{\mu}{\mu_0}$$

- Continuity equation.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 = \frac{\partial \rho}{\partial t} + \frac{d}{dx_j} (\rho v_j) \quad * \quad \frac{L/v_0}{\rho_0}$$

$$\begin{aligned} \frac{\frac{\partial \rho}{\partial t}}{\frac{\partial t}{L/v_0}} + L \nabla \cdot \frac{\rho}{\rho_0} \frac{\vec{v}}{v_0} &= \frac{\partial \rho^*}{\partial t^*} + \nabla^* \cdot (\rho^* \vec{v}^*) \\ &= \frac{\partial \rho^*}{\partial t^*} + \frac{d}{dx_j^*} (\rho^* v_j^*) = 0 \end{aligned}$$

- No dimensionless parameters.

- Continuity is not influenced directly by any flow parameter.

- Momentum equation. high speed flow,
 $\rho \vec{g}$ is negligible.

$$\rho \frac{D \vec{v}}{Dt} = -\nabla p + \cancel{\rho \vec{g}} + \nabla \cdot \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] \quad * \frac{4 \nu_0}{\rho_0 \nu_0}$$

$$\frac{\rho}{\rho_0} \frac{D \vec{v}/\nu_0}{D t/\nu_0} = -L \nabla \frac{p/p_0}{\rho \nu_0^2} + \frac{\mu}{L \nu_0 \rho_0} L \nabla \cdot \left[\frac{\partial u_i/\nu_0}{\partial x_j/L} + \frac{\partial u_j/\nu_0}{\partial x_i/L} \right]$$

$$\rho^* \frac{D u_i^*}{D t^*} = -\nabla^* p^* + \boxed{\frac{1}{Re}} \nabla^* \cdot \left[\nabla^* u^* + (\nabla^* u^*)^T \right]$$

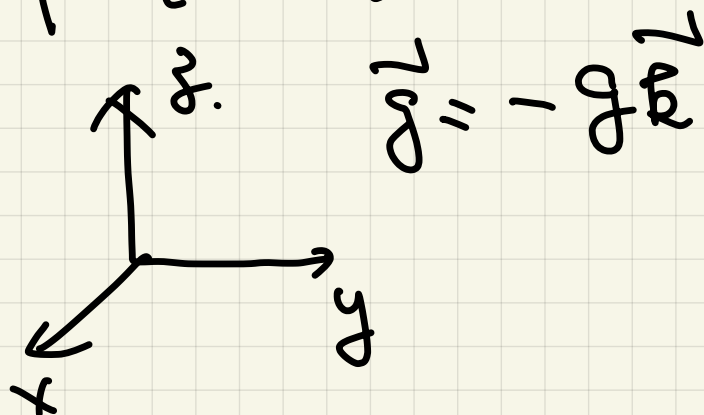
$$\rho^* \frac{D u_i^*}{D t^*} + \frac{\partial}{\partial x_j^*} (\rho^* u_j^* u_i^*) = -\frac{\partial p^*}{\partial x_i^*} + \frac{1}{Re} \frac{\partial}{\partial x_j^*} \left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*} \right)$$

- For low speed flow, $\rho \vec{g}$ considered.

- For small density change; $\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p$

Boussinesq approximation for ideal gas.

$$p = p_0 + \Delta p \approx p_0 (1 - \beta \Delta T) \quad | \quad p = \rho R T$$



$$\vec{g} = -g \hat{e}_z$$

$$\rho = \frac{p}{RT} \quad \frac{\partial \rho}{\partial T} = -\frac{p}{RT^2}$$

$$-\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p = \frac{p}{RT^2} \frac{RT}{p} = \frac{1}{T}$$

$$P = P_0 + \Delta P$$

$$(P_0 + \Delta P) \frac{D\vec{V}}{Dt} = -\nabla P + \rho_0 \vec{g} - \rho_0 \beta \Delta T \vec{g} + \nabla \cdot \vec{\tau}^*$$

$$\rho_0 \frac{D\vec{V}}{Dt} = -\nabla (P - P_0 + \rho_0 g z) - \rho_0 \beta (T - T_0) \vec{g} + \nabla \cdot \vec{\tau}^*$$

$$\beta^* = \frac{\beta}{\beta_0}, \quad P^* = \frac{P - P_0 + \rho_0 g z}{\rho_0 V_0^2}$$

$$\left[\rho_0 \frac{D\vec{V}}{Dt} = -\nabla (P - P_0 + \rho_0 g z) - \rho_0 \beta (T - T_0) \vec{g} + \nabla \cdot \vec{\tau}^* \right]$$

$$\frac{D\vec{V}^*}{Dt} = -\nabla^* P^* - \frac{\rho_0 L}{\rho_0 V_0^2} \beta (T - T_0) \vec{g} + \frac{1}{Re} \nabla^* \cdot \vec{\tau}^*$$

$\frac{L/V_0}{\rho_0 V_0}$

$$\frac{\beta}{\beta_0} \frac{(T - T_0)}{T_w - T_0} \frac{\vec{g}}{g} \quad \boxed{\frac{\rho_0^2 L^3 g \beta_0 (T_w - T_0)}{\mu_0^2}} \quad \boxed{\frac{1}{\rho_0^2 L^2 V_0^2 / \mu_0^2}}$$

Grashoff number $\frac{1}{Re^2}$

$$\frac{\partial u_x^*}{\partial t^*} + \frac{\partial u_y^* u_x^*}{\partial x_y^*} = -\frac{\partial P^*}{\partial x_x^*} + \frac{1}{Re} \frac{\partial}{\partial x_y^*} \left(\frac{\partial u_x^*}{\partial x_y^*} + \frac{\partial u_y^*}{\partial x_x^*} \right)$$

$$+ \frac{Gr}{Re^2} \beta^* T^* \vec{g}^* \quad \text{Buoyancy force}$$

- The Grashoff number is a measure of buoyancy or free convective effect.

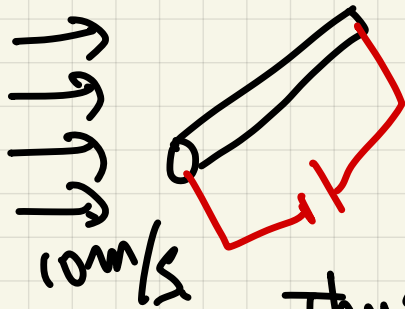
- If there is a free stream.

$$\frac{Gr}{Re^2} = \beta(T_w - T_\infty) \frac{gL}{V_0^2} = \beta(T_w - T_\infty) \frac{1}{V_0^2/gL}$$

$$= \frac{\beta(T_w - T_\infty)}{Fr} \propto \frac{1}{Fr}$$

- For example,
a hot wire anemometer

$d_i = 0.1 \text{ mm}$.



$$Fr = \frac{V_0^2}{gL} = \frac{10^2}{9.8 \times 10^{-4}} \approx 10^5$$

change of resistance
 $R \rightarrow I \cdot \Delta V$
 $V \rightarrow I$

Thus, the effect of buoyancy is negligible.

- If there is no free stream, the fluid motion is due to temperature difference - natural convection

reference velocity $\frac{\mu_0}{\rho_0 L}$

$$Re = \frac{\rho_0 V_0 L}{\mu_0} = \rho_0 \frac{\mu_0}{\rho_0 L} \cdot L = 1$$

$$\therefore \frac{Gr}{Re^2} = Gr.$$

- Energy equation.

$$\frac{1}{\rho_0 C_p} \frac{1}{T_w - T_0} \frac{1}{V_0}$$

$$\rho C_p \frac{DT}{Dt} = \frac{DP}{Dt} + \nabla \cdot (k \nabla T) + \mu_0 \left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial x} \right) \frac{\partial u}{\partial y}$$

$$\rho^* C_p^* \frac{DT^*}{Dt^*} = \underbrace{\frac{V_0^2}{C_p (T_w - T_0)}}_{Ec} \frac{DP^*}{Dt^*} + \underbrace{\frac{\rho_0 V_0 L}{\mu_0}}_{Re} \underbrace{\frac{\mu_0 C_p}{k_0}}_{Pr} \nabla^* \cdot \nabla^* T^* + \underbrace{\frac{V_0^2}{C_p (T_w - T_0)}}_{Ec} \underbrace{\frac{1}{\rho_0 V_0 L / \mu_0}}_{Re} \underbrace{\left(\frac{\partial u^*}{\partial y^*} + \frac{\partial u^*}{\partial x^*} \right) \frac{\partial u^*}{\partial y^*}}_{\Phi^*}$$

$$\rho^* C_p^* \frac{DT^*}{Dt^*} = Ec \frac{DP^*}{Dt^*} + \frac{1}{Re Pr} \nabla^* \cdot (\nabla^* T^*) + \frac{Ec}{Re} \Phi^*$$

$$Re = \frac{\rho V_0 L}{\mu_0}, \quad Pr = \frac{\mu_0 C_p}{k_0}, \quad Ec = \frac{V_0^2}{C_p (T_w - T_0)}$$

$$Re \cdot Pr = \text{Peclet number} = \frac{\rho V_0 L}{k_0 / C_p}$$

For low speed flow, both the pressure and dissipation terms are negligible.

2-10. Vorticity in incompressible flow.

$$\vec{\omega} = \nabla \times \vec{v}, \quad g = -g\hat{k}. \quad \begin{array}{c} z \\ \nearrow \\ x \quad y \end{array}$$

$$\frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}$$

$$\rho \frac{D\vec{v}}{Dt} = -\nabla p - \rho g\hat{k} + \mu \nabla^2 \vec{v}$$

$$(\vec{v} \cdot \nabla) \vec{v} = \nabla \frac{v^2}{2} - \vec{v} \times \vec{\omega}$$

$$\nabla^2 \vec{v} = \nabla(\nabla \cdot \vec{v}) - \nabla \times \vec{\omega} = -\nabla \times \vec{\omega}$$

$$\therefore \rho \frac{D\vec{v}}{Dt} + \nabla \frac{v^2}{2} + \nabla p + \nabla(\rho g z) = \rho \vec{v} \times \vec{\omega} - \mu \nabla \times \vec{\omega}$$

$$\rho \frac{D\vec{v}}{Dt} + \nabla \left(p + \frac{v^2}{2} + \rho g z \right) = \rho \vec{v} \times \vec{\omega} - \mu \nabla \times \vec{\omega}$$

- For irrotational flow.

$\vec{\omega} = \nabla \times \vec{v} = 0$. $\vec{v} = \nabla \phi$, where ϕ is the velocity potential

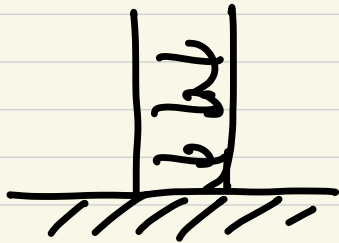
$$\nabla \left(\rho \frac{d\phi}{dt} + p + \frac{v^2}{2} + \rho g z \right) = 0$$

$$\rho \frac{d\phi}{dt} + p + \frac{v^2}{2} + \rho g z = C$$

- Bernoulli eq $p + \frac{v^2}{2} + \rho g z = C$ along a streamline.

- For irrotational flow.

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0. \quad \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y}$$



at the wall $u=0$

$$\frac{\partial u}{\partial y} = 0.$$

Therefore, the velocity slips at the boundary of irrotational flow.

- Take curl of momentum equation.

$$\nabla \times \left(\rho \frac{d\vec{v}}{dt} + \vec{v} \cdot \nabla \vec{v} = -\nabla p + \mu \nabla^2 \vec{v} \right)$$

$$\nabla \times (\vec{v} \cdot \nabla \vec{v}) = \vec{v} \cdot \nabla (\nabla \times \vec{v}) - (\nabla \times \vec{v}) \cdot \nabla \vec{v} + \underbrace{(\nabla \cdot \vec{v}) \nabla \times \vec{v}}_{\text{incompressible } 0}$$

$$\rho \frac{d\vec{\omega}}{dt} + \rho \vec{v} \cdot \nabla \vec{\omega} - \rho \vec{\omega} \cdot \nabla \vec{v} = \mu \nabla^2 \vec{\omega}$$

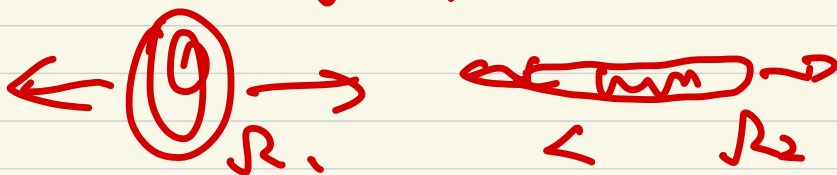
$$\rho \frac{d\vec{\omega}}{dt} = \rho \vec{\omega} \cdot \nabla \vec{v} + \mu \nabla^2 \vec{\omega} \quad \leftarrow \text{Helmholtz equation for hydrodynamics.}$$

$$\rho \omega_x \frac{\partial \vec{v}}{\partial x} + \rho \omega_y \frac{\partial \vec{v}}{\partial y} + \rho \omega_z \frac{\partial \vec{v}}{\partial z} \Rightarrow \text{for 2-D flow}$$

vortex stretching

$$\rho \vec{\omega} \cdot \nabla \vec{v} \rightarrow 0$$

$$\omega_z = 0$$



2-1 Two-dimensional consideration the stream function.

- For constant ρ and μ .

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \delta x - \rho \frac{\partial \phi}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \delta y - \rho \frac{\partial \phi}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)$$

- stream function.

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}.$$

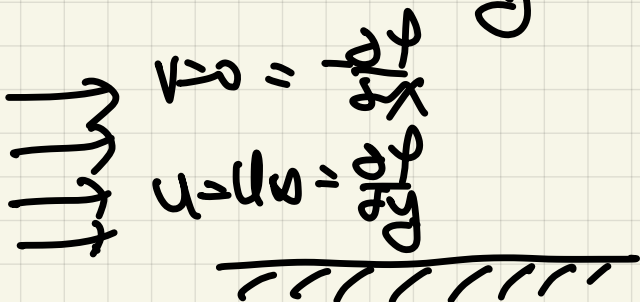
$$\nabla^2 \phi = -\omega_z \rightarrow 0.$$

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} = -\nabla^2 \phi$$

$$\frac{\partial \omega_z}{\partial t} + u \frac{\partial \omega_z}{\partial x} + v \frac{\partial \omega_z}{\partial y} = \nu \left(\frac{\partial^2 \omega_z}{\partial x^2} + \frac{\partial^2 \omega_z}{\partial y^2} \right)$$

vortex stretching is zero for 2-D flow.

$$\therefore \frac{\partial}{\partial t}(\nabla^2 \phi) + \frac{\partial \phi}{\partial y} \frac{\partial}{\partial x}(\nabla^2 \phi) - \frac{\partial \phi}{\partial x} \frac{\partial}{\partial y}(\nabla^2 \phi) = \nu \nabla^2 \nabla^2 \phi$$



$$v=0 = \frac{\partial \psi}{\partial x}$$

$$u=0 = \frac{\partial \psi}{\partial y}$$

$$u=0 = \frac{\partial \psi}{\partial y}$$

$$v=0 = \frac{\partial \psi}{\partial x}$$

- Non-dimensional form.

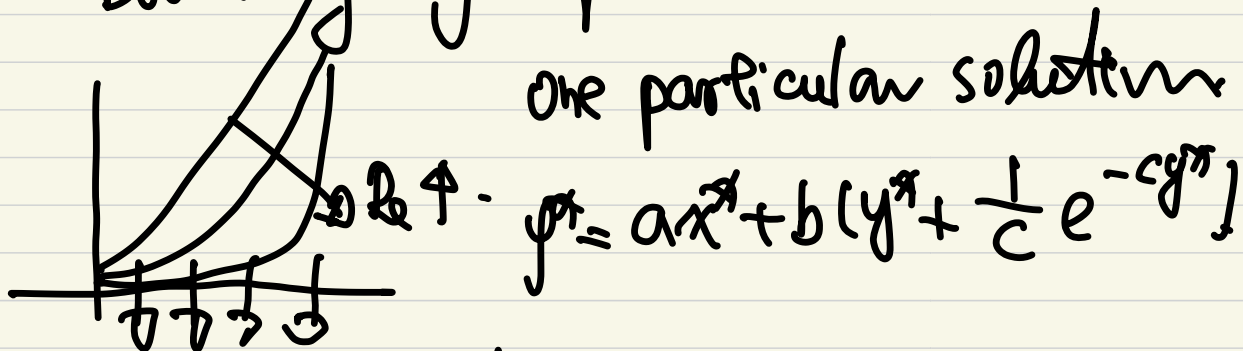
$$t^* = t / \tau_0, \quad x^* = \frac{x}{L}, \quad y^* = \frac{y}{V_0 L}$$

$$\frac{\partial}{\partial t^*} (\tau^2 \psi^*) + \frac{\partial \psi^*}{\partial y^*} \frac{\partial}{\partial x^*} (\tau^2 \psi^*) - \frac{\partial \psi^*}{\partial x^*} \frac{\partial}{\partial y^*} (\tau^2 \psi^*) = \frac{1}{Re} \tau^4 \psi^*, \quad Re = \frac{V_0 L}{\nu}$$

- Creeping flow $Re \ll 1$.

$$\nabla^2 \psi^* \approx 0. \quad \sim \nabla p = \mu \nabla^2 \mathbf{v}$$

- Boundary layer flow $Re \gg 1$.



Then the equation becomes,

$$\frac{\partial}{\partial t^*} (\tau^2 \psi^*) + \frac{\partial \psi^*}{\partial y^*} \frac{\partial}{\partial x^*} (\tau^2 \psi^*) - \frac{\partial \psi^*}{\partial x^*} \frac{\partial}{\partial y^*} (\tau^2 \psi^*) = \frac{1}{Re} \tau^4 \psi^*$$

$$0 + 0 - a(-b c^2 e^{-c y^*}) = \frac{1}{Re} b c^3 e^{-c y^*}$$

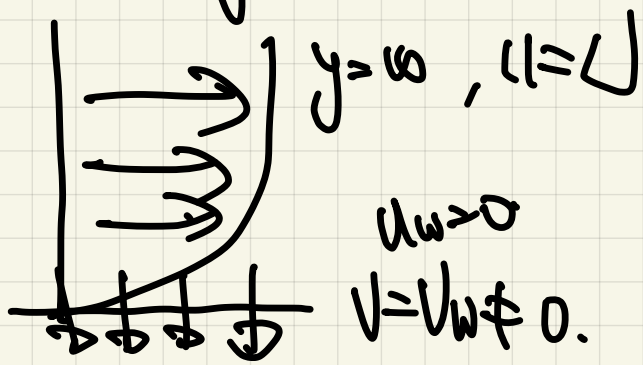
Therefore $c = a Re$

$$u^* = \frac{\partial \psi^*}{\partial y^*} = b(1 - e^{-c y^*})$$

$$v^* = - \frac{\partial \psi^*}{\partial x^*} = -a$$

$\rightarrow$ flow with suction

- uniform suction on a plane



$$\rho v_w \frac{\partial u}{\partial y} = \mu \frac{\partial^2 u}{\partial y^2}$$

↑
constant.

$$\int (\rho v_w \frac{\partial u}{\partial y} = \mu \frac{\partial^2 u}{\partial y^2}) dy$$

$$\rho v_w u = \mu \frac{\partial u}{\partial y} + C \quad \text{at } y=\infty, u=U, \frac{\partial u}{\partial y} \rightarrow 0$$

$$C = \rho v_w U.$$

$$\rho v_w (u - U) = \mu \frac{\partial u}{\partial y}$$

$$dy = \frac{\mu}{\rho v_w} \frac{d(u - U)}{u - U} = \frac{\mu}{\rho v_w} \frac{d(U - u)}{U - u}$$

$$y = \frac{\mu}{\rho v_w} \ln(U - u) + B.$$

$$y=0, u=0 \Rightarrow 0 = \frac{\mu}{\rho v_w} \ln U + B \Rightarrow B = -\frac{\mu}{\rho v_w} \ln U$$

$$y = \frac{\mu}{\rho v_w} \ln(U - u) - \frac{\mu}{\rho v_w} \ln U.$$

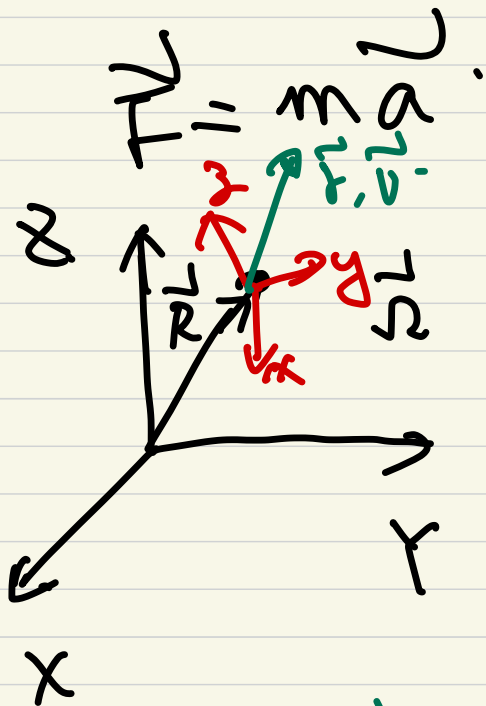
$$\frac{\rho v_w y}{\mu} = \ln(U - u) - \ln U = \ln \frac{U - u}{U}.$$

$$e^{\frac{\rho v_w y}{\mu}} = \frac{U - u}{U} \Rightarrow U - u = U e^{-\frac{\rho v_w y}{\mu}}$$

$$u = U \left(1 - e^{-\frac{\rho v_w y}{\mu}} \right)$$

v_w is negative

2-12. Non-inertia coordinate system



X, Y, Z : inertia frame
 x, y, z : non-inertia frame.
 translating and rotating
 $\vec{R}$ $\vec{\omega}$
 relative to the inertia
 frame.

$\vec{r}$ and $\vec{v}$ at the non-inertia frame.

Then,

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} + \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) + \frac{d\vec{v}}{dt} + 2\vec{\omega} \times \vec{v}$$